

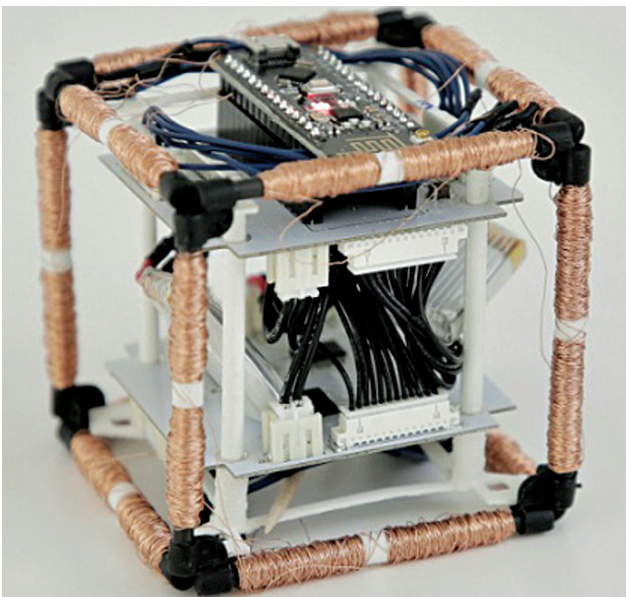


The enzyme-free sensor is more robust and not temperature sensitive, so could be particularly useful in hot countries or where refrigeration is limited (Credit: www.bath.ac.uk)

sensor instead of being enzyme-based, the new technology is robust, has a long shelf-life, and can be tuned to detect lower glucose concentrations than current systems.

Scalable and reconfigurable robots using electromagnets

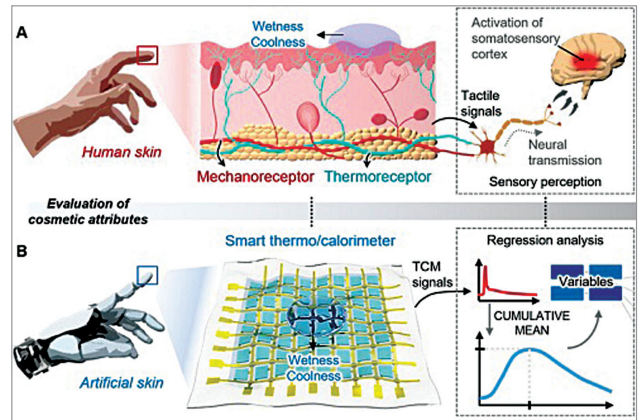
An engineering team from MIT CSAIL has devised a cube-shaped reconfigurable robot structure called ElectroVoxel that utilises an electromagnet based actuation framework to reconfigure in three dimensions via pivoting. The modular self-reconfigurable robot (MSRR) has electromagnets lined at the edges of each cube that attract or repel each other to reconfigure into different shapes and sizes. Powered by a LiPo battery, the 3D printed cube-shaped structure has nearly 800 turns of electromagnet wound around a ferromagnetic core. Applications include space exploration, search and rescue, shape-changing user interfaces, and endlessly more.



Elements of an electromagnetically actuated reconfigurable robot (Credit: MIT CSAIL)

Intelligent tactile sensor that measures skin sensations

Amorepacific, in collaboration with the Ulsan National Institute of Science and Technology (UNIST), has developed an intelligent tactile sensor that can measure skin



The intelligent tactile sensor co-developed by Amorepacific and UNIST (Credit: PR Newswire)

sensations by combining machine learning with measurement technology. By withstanding external pressure and staying intact, the ultra-thin flexible sensor can recognise and measure the temperature, moisture levels, and the type of solution used on the skin surface. The new sensor will enable the development of finetuned cosmetics and skin products.

Used face masks turned into batteries

Scientists at NUST MISIS, together with their foreign colleagues, have developed a new technology for producing



Tonnes of used face masks to be turned into energy (Credit: <https://en.misis.ru>)